Handling Header & Item Data in ABAP

Using FOR ALL ENTRIES Effectively

Problem Statement:

Requirement:

- Fetch Order Header data.
- Fetch corresponding Item data.
- Combine both into a single output.

Challenge:

- Data exists in **two different tables**.

Code:

```
*&---------------------------------------------------------------------
*& Report ZP34_CR_FOR_ALL_ENTRIES
*&---------------------------------------------------------------------
*&
*&---------------------------------------------------------------------
REPORT zp34_cr_for_all_entries.

DATA : lv_ono TYPE zordernum_pl.
DATA : lv_flag TYPE boolean.

TYPES : BEGIN OF lstr_tab1,
          order_number TYPE zordernum_pl,
          order_date   TYPE zorderdate_pl,
          payment_mode TYPE zpayment_pl,
          currency     TYPE zcurrency_pl,
        END OF lstr_tab1.

DATA : lit_tab1 TYPE TABLE OF lstr_tab1.
DATA : lwa_tab1 TYPE lstr_tab1.

TYPES : BEGIN OF lstr_tab2,
          order_number  TYPE zordernum_pl,
          order_itm_num TYPE zorderitmno_pl,
          item_cost     TYPE zitemcost_pl,
        END OF lstr_tab2.

```

```
DATA : lit_tab2 TYPE TABLE OF lstr_tab2.
DATA : lwa_tab2 TYPE lstr_tab2.

TYPES : BEGIN OF lfinal,
          order_number  TYPE zordernum_pl,
          order_date    TYPE zorderdate_pl,
          payment_mode  TYPE zpayment_pl,
          currency      TYPE zcurrency_pl,
          order_itm_num TYPE zorderitmno_pl,
          item_cost     TYPE zitemcost_pl,
        END OF lfinal.

DATA : lit_final TYPE TABLE OF lfinal.
DATA : lwa_final TYPE lfinal.

SELECT-OPTIONS : s_ono FOR lv_ono.

SELECT order_number order_date payment_mode currency
FROM zohtable_pl
INTO TABLE lit_tab1
WHERE order_number IN s_ono
ORDER BY order_number.
```

```
SELECT order_number order_date payment_mode currency
FROM zohtable_pl
INTO TABLE lit_tabl
WHERE order_number IN s_ono
ORDER BY order_number.

IF lit_tabl IS NOT INITIAL." NOT INITIAL - not null
  SELECT order_number order_itm_num item_cost
  FROM zoitmtable_pl
  INTO TABLE lit_tab2
  FOR ALL ENTRIES IN lit_tabl
  WHERE order_number = lit_tabl-order_number.
ENDIF.
```

```
LOOP AT lit_tabl INTO lwa_tabl.
  LOOP AT lit_tab2 INTO lwa_tab2 WHERE order_number = lwa_tabl-order_number.
    lwa_final-order_number = lwa_tabl-order_number.
    lwa_final-order_date = lwa_tabl-order_date.
    lwa_final-payment_mode = lwa_tabl-payment_mode.
    lwa_final-currency = lwa_tabl-currency.
    lwa_final-order_itm_num = lwa_tab2-order_itm_num.
    lwa_final-item_cost = lwa_tab2-item_cost.
    APPEND lwa_final TO lit_final.
    CLEAR : lwa_final.
    lv_flag = 'X'.
  ENDLOOP.
  IF lv_flag = ' '.
    lwa_final-order_number = lwa_tabl-order_number.
    lwa_final-order_date = lwa_tabl-order_date.
    lwa_final-payment_mode = lwa_tabl-payment_mode.
    lwa_final-currency = lwa_tabl-currency.
    APPEND lwa_final TO lit_final.
    CLEAR : lwa_final.
  ENDIF.
  CLEAR : lv_flag.
ENDLOOP.

LOOP AT lit_final INTO lwa_final.
  WRITE : / lwa_final-order_number, lwa_final-order_itm_num, lwa_final-order_date,
            lwa_final-payment_mode, lwa_final-item_cost, lwa_final-currency.
ENDLOOP.
```

Step-by-Step Explanation of Code:

1) Header Data Fetch:

Code :

```
SELECT order_number order_date payment_mode currency
FROM zohtable_p1
INTO TABLE lit_tab1
WHERE order_number IN s_ono.
```

Result :

- All matching header records stored in internal table.

2) FOR ALL ENTRIES:

Code:

```
IF lit_tab1 IS NOT INITIAL.
SELECT order_number order_itm_num item_cost
FROM zoitmtable_p1
INTO TABLE lit_tab2
FOR ALL ENTRIES IN lit_tab1
WHERE order_number = lit_tab1-order_number.
ENDIF.
```

Key Concept:

- Fetches item data for all header entries.

Critical Rule:

- If lit_tab1 is empty → full table fetch (performance issue).

Why Not SELECT Inside LOOP?

Bad Practice:

- SELECT inside LOOP → multiple DB hits.

Better Approach:

- FOR ALL ENTRIES → single DB hit.

➔ Performance optimized design.

3) Data Merging Logic:

Code :

```
LOOP AT lit_tab1 INTO lwa_tab1.
  LOOP AT lit_tab2 INTO lwa_tab2
    WHERE order_number = lwa_tab1-order_number.
```

Concept:

- Manual join in ABAP layer.
- Combines header + item data.

4) Building Final Structure:

Code :

```
APPEND lwa_final TO lit_final.
```

Why final table?

- Unified dataset
- Ready for display / ALV

Real-time:

- Used before ALV reports or output formatting.

5) Handling Missing Items:

Code :

```
IF lv_flag = ' '.
```

Insight:

- Handles cases where:
  - Header exists
  - No item records found

Ensures:

- No data loss.
- Complete reporting.

6) Output Display:

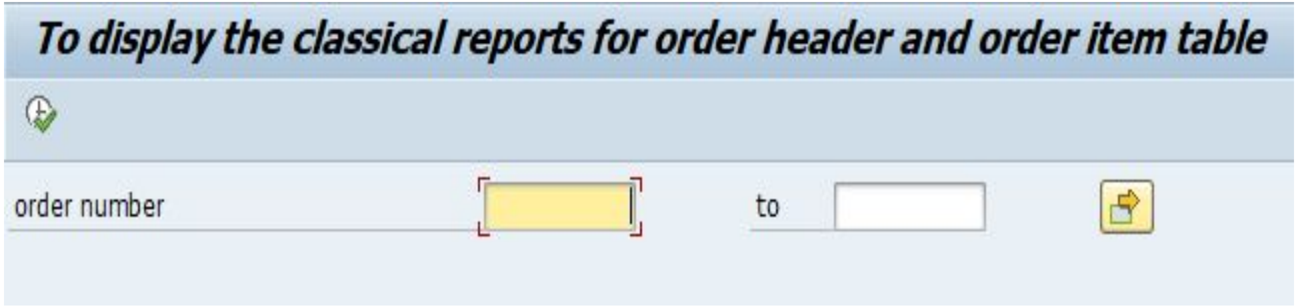
Code :

```
LOOP AT lit_final INTO lwa_final.
WRITE : / ...
ENDLOOP.
```

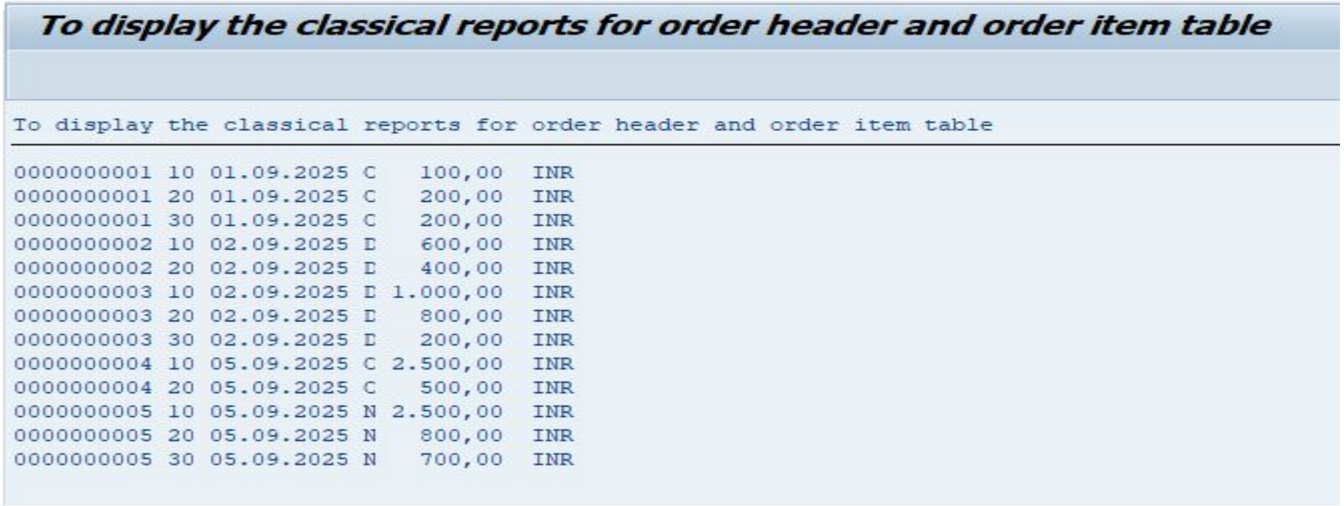
Final step:

- Displays merged dataset.

Selection Screen:



Output Screen:



Key Takeaways:

- FOR ALL ENTRIES reduces DB calls.
- Always check NOT INITIAL.
- Manual merging gives flexibility.
- Avoid SELECT inside LOOP.